

Applied Algebra

Spring 2026

Instructor: Tuan Tran

Exercise Sheet 4

Deadline: by the end of next Tuesday

Instructions. You are encouraged to work in pairs. When submitting your solutions, list the team members and circle the name of the person responsible for writing up the solutions that week.

Please submit written solutions to **TWO** of the following problems.

Problem 1. Let $n = 2^6 \cdot 3^4 \cdot 5^3$.

- (i) How many positive divisors does n have?
- (ii) How many of them are perfect squares (i.e., of the form k^2 for some integer k)?
- (iii) How many of them are divisible by $2^2 \cdot 3$?
- (iv) How many of them are relatively prime to 30?

Problem 2. Find

- (i) $3^{100} \bmod 10$,
- (ii) $7^{222} \bmod 40$,
- (iii) $6^{50} \bmod 15$,
- (iv) $8^{123} \bmod 20$.

Problem 3. Solve the following congruences.

- (i) $6x \equiv 15 \pmod{21}$,
- (ii) $8x \equiv 7 \pmod{20}$.
- (iii) $x \equiv 1 \pmod{4}$, $x \equiv 3 \pmod{7}$.
- (iv) $x \equiv 2 \pmod{4}$, $x \equiv 3 \pmod{6}$.

Problem 4. Let a, b be positive integers with $\gcd(a, b) = 1$.

- (i) Show that for every integer n , there exists an integer x such that

$$n \equiv ax \pmod{b}.$$

- (ii) Deduce that every integer $n \geq (a-1)(b-1)$ can be written in the form

$$n = ax + by$$

for some nonnegative integers x, y .